

School of Social Sciences

COURSE OUTLINE

ECO 403: Advanced Quantitative Economics

Contact Hours: 4 Hours per Week

Credit Hours: 4

Prerequisite: ECO 202: Intermediate Macroeconomics

Course Description

This course introduces advanced analytical and quantitative methods used in modern economic theory and empirical research. Topics include dynamic optimization, optimal control theory, general equilibrium analysis, game theory, and econometric modeling. Emphasis is placed on mathematical formulation, strategic economic interactions, and data-driven policy evaluation. Students also explore computational techniques and simulations for solving complex economic models.

Course Objectives

By the end of the course, the student should be able to:

- Apply dynamic optimization and optimal control techniques to economic decision-making problems.
- Analyze general equilibrium models and welfare implications in modern economies.
- Evaluate strategic interactions and market structures using game theory concepts.
- Utilize econometric methods to test economic theories and interpret empirical data.
- Develop computational and quantitative skills for modeling and solving advanced economic problems.

Course Content Outline

Week	Topic	Subtopics
1	Introduction to Quantitative Economics	Review of intermediate microeconomics, macroeconomics, mathematical methods, and statistical foundations
2	Dynamic Optimization I	Intertemporal choice, constrained optimization, Lagrange multipliers, economic applications
3	Dynamic Optimization II	Dynamic programming, Bellman equations, value functions, recursive methods
4	Optimal Control Theory	Hamiltonians, Pontryagin's principle, continuous-time optimization models
5	Representative Agent Models	Consumption-saving decisions, labor supply, equilibrium behavior in representative economies
6	General Equilibrium Analysis	Competitive equilibrium, welfare theorems, Pareto efficiency, market clearing
7	Overlapping Generations Models	Two-period models, capital accumulation, intergenerational welfare analysis
8	Welfare Economics and Policy	Social welfare functions, redistribution, taxation, efficiency-equity tradeoffs
9	Game Theory I	Strategic form games, dominant strategies, mixed strategies, Nash equilibrium
10	Game Theory II	Oligopoly models, repeated games, bargaining, information asymmetry
11	Econometric Modeling I	Regression analysis, causal inference, model specification, hypothesis testing
12	Econometric Modeling II	Regression discontinuity design, bunching estimators, policy evaluation methods
13	Computational Economics	MATLAB-based simulations, numerical methods, calibration of economic models
14	Advanced Applications and Review	Integrated quantitative analysis, empirical case studies, review of modern economic research methods

Teaching Methodology

- Interactive lectures and guided mathematical demonstrations.
- Problem-solving sessions involving quantitative economic models and case studies.
- Computer-based simulations and econometric analysis using statistical software such as MATLAB.
- Group discussions and presentations on theoretical and empirical economic applications.
- Continuous assessment through assignments, tutorials, and analytical projects.

Assessment

Component	Weight
Continuous Assessment Tests (CATs)	30%
Assignments and Quizzes	10%
Final Examination	60%

Reference Materials

- Recursive Macroeconomic Theory – Lars Ljungqvist & Thomas J. Sargent, MIT Press.
- Dynamic Economics: Quantitative Methods and Applications – Jérôme Adda & Russell Cooper, MIT Press.
- Microeconomic Theory – Andreu Mas-Colell, Michael D. Whinston & Jerry R. Green, Oxford University Press.
- Game Theory – Drew Fudenberg & Jean Tirole, MIT Press.
- Mostly Harmless Econometrics – Joshua D. Angrist & Jörn-Steffen Pischke, Princeton University Press.
- Econometric Analysis – William H. Greene, Pearson Education.
- Mathematics for Economists – Carl P. Simon & Lawrence Blume, W.W. Norton & Company.
- Fundamental Methods of Mathematical Economics – Alpha C. Chiang & Kevin Wainwright, McGraw-Hill Education.